



Investigating the penultimate and last glacial cycles of the Süttő loess section (Hungary) using luminescence dating, high-resolution grain size, and magnetic susceptibility data

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ABSTRACT

The loess–paleosol record at Süttő provides an excellent high-resolution archive of climate and environmental changes. Loess deposits with a thickness of 20 m cover the travertine at Süttő, in the northern part of Hungary, next to the right bank of the Danube River. The loess sequence is intercalated with two greyish stratified horizons, three brownish steppe-like soils and a pedocomplex, including a reddish-brown paleosol covered by a chernozem-like paleosol. Detailed infrared stimulated luminescence (IRSL) dating was carried out, revealing more or less continuous sedimentation from Marine Isotope Stage (MIS) 6–MIS 2. Sedimentological and paleomagnetic investigations have been performed to provide a high-resolution study of the Süttő loess–paleosol sequence, in order to reconstruct the paleoclimate and environmental changes during the penultimate and last glacial cycles. Grain size and magnetic susceptibility (MS) records of the Süttő profile basically show similar patterns to those of other European loess sections. However, small differences are traced at Süttő, including more intensive deposition of coarser grained materials during MIS 6–4. The loess record at Süttő has a transitional position within the European loess belt, as its sediment was deposited under an intermediate climate between the loess successions from the southern Carpathian Basin, which accumulated under drier climate, and the loess sequences from Western Europe, which developed under wetter climate during the Middle and Late Pleniglacial.

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1. Introduction and regional setting

Loess and loess-like sediments cover more than 30% of the surface of Hungary. In the Transdanubian Hills, Gödöllő Hills and on river terraces along the right bank of the River Danube, loess deposits are widespread. The loess–paleosol sequence at Süttő is located in northern Hungary, close to the right bank of the River Danube (47°44.26' N, 18°26.87' E) (Fig. 1a). The section under study is located about 35 km from the Middle Pleistocene key loess section of Basaharc (Frechen et al., 1997; Bajnóczi et al., 2006) and situated on a Danube-terrace, which is covered by travertine (Fig. 1b). The overlying loess deposits are up to 20 m thick, and contain two greyish, laminated horizons, three brownish steppe-like soils and a thick pedocomplex including a dark brown

chernozem-like paleosol and a reddish-brown paleosol (Fig. 2). The geological setting of the section under study is discussed in more detail by Novothny et al. (2009).

Parent material of windblown sediments derives mostly from the alluvial plains of the Danube-system. The alluvial silt- and sand-sized material was deflated and accumulated in adjacent areas mostly after short-distance transport. There are three possible sources of the loess-sandy loess deposited around Süttő:

1. Glacial material from the Alpine region carried by the paleo-Danube (Smalley and Leach, 1978)
2. Weathered material of the flysch-type rocks of the northern Carpathians delivered by the paleo-tributaries of the Danube (Smalley and Leach, 1978)
3. Local sources (Bikol Creek) also made some contribution (Pécsi and Richter, 1996).

Independent age control is provided by uranium-series ($^{230}\text{Th}/^{234}\text{U}$) dating of the travertine beneath the loess–paleosol sequence, yielding mid-Pleistocene ages, ranging from 235 ± 21 to

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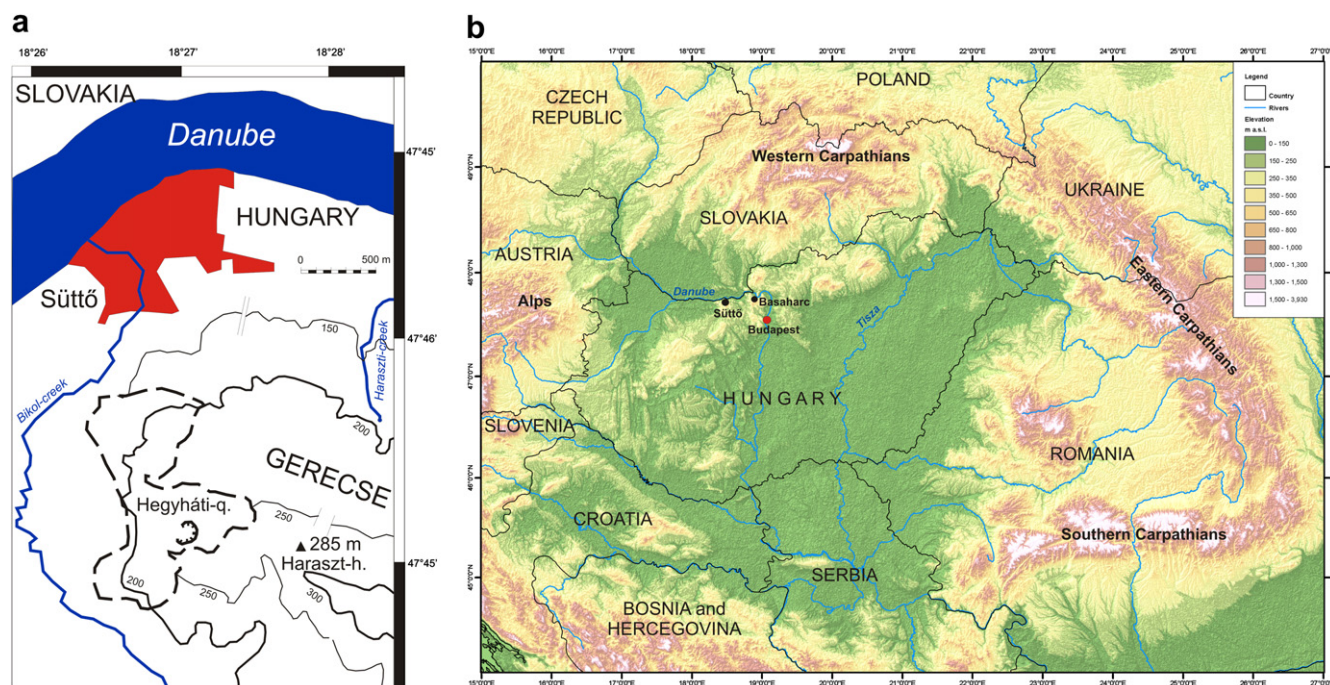


Fig. 1. a) Map showing the location of the Süttő section in northern Hungary. b) Map showing the location of the Carpathian Basin.

314 ± 45 ka (Sierralta et al., 2010) indicating a correlation to Marine Isotope Stage (MIS) 7–MIS 9. Detailed chronological investigations have been accomplished on the loess sequence, comprising infrared stimulated luminescence (IRSL), infrared radiofluorescence (IRRF) measurements from loess samples, amino acid racemization (AAR) from molluscs and radiocarbon dating obtained from molluscs and charcoal samples (Novothny et al., 2009, 2010). The available chronological dataset revealed a detailed Late Pleistocene record for the loess–paleosol succession from Süttő, providing an excellent climate archive for the time period ranging from MIS 6 to MIS 2. Based on the established chronological framework, high-resolution investigations have been carried out, including granulometry, environmental magnetism, pedology, malacozoology, geochemistry by means of organic carbon, stable isotopes and biomarkers using *n*-alkanes to reconstruct paleoclimate and paleoenvironmental changes. This study focuses on the presentation and interpretation of granulometry and magnetic susceptibility results complemented with further IRSL dating to achieve finer chronology of the sediments deposited in MIS 5 and 4. As the grain-size distribution of loess is a function of climate, the aim is to reconstruct some aspects of the paleoclimate evolution.

2. Methods

2.1. Samples

Two quasi-continuous vertical trenches were prepared for sampling. The lower part of the loess sequence was opened up on the N–NW wall of the quarry, while the upper part was explored on the freshly excavated SW wall, as it was found to have a more detailed record for the upper part of the sequence. The N–NW wall is identical with the previously described section at Süttő in Novothny et al. (2009, 2010). The vertical trenches overlapped for more than 2 m for better comparison and correlation of the sediment units between the two walls.

After careful cleaning and detailed description of the sediment record, bulk samples for grain-size analysis were collected in 2 cm intervals, resulting in more than 1000 samples taken from the whole sequence. For magnetic susceptibility measurements samples were collected from the same positions, but in 10 cm intervals. Where sampling with cubes was not possible, samples were taken in overlapping blocks. Altogether, 400 small cubes and 180 soft-core samples were collected using a modified sampling technique developed by Schnepf et al. (2008). Eleven additional samples (LUM 1595–1604 and 2077) were collected for luminescence dating by pushing metal tubes into the previously cleaned loess wall.

2.2. Grain-size analysis

Grain-size distributions of the granulometry samples were determined using a Beckman–Coulter LS 13320 PIDS laser diffraction particle size analyser. It covers a wide range of grain sizes (0.02–2000 µm) and requires only a small amount of sample material, but its most important advantage is the short analysis time. However, compared to the sieve-pipette method, some disadvantages should also be mentioned, including the problematic measurement of organic materials, which may absorb light, or the underestimation of the clay content of the samples, due to the platy shape of the clay minerals. Therefore, a unique relationship between the results of a laser diffraction particle size analyser and those of the classic sieve-pipette method cannot be achieved. However this problem is more pronounced in clay-rich than in sand-rich samples (Beuselinck et al., 1998). The study followed the routine of Machalett et al. (2008) for sample preparation, rejecting removal of organic material and carbonates as they showed a negligible contribution to the final results during the test measurements, and as secondary carbonates could not be separated from the primary carbonates. Similar results regarding organic material were reported by Beuselinck et al. (1998). Intensive ultrasonic disaggregation was also avoided and replaced by

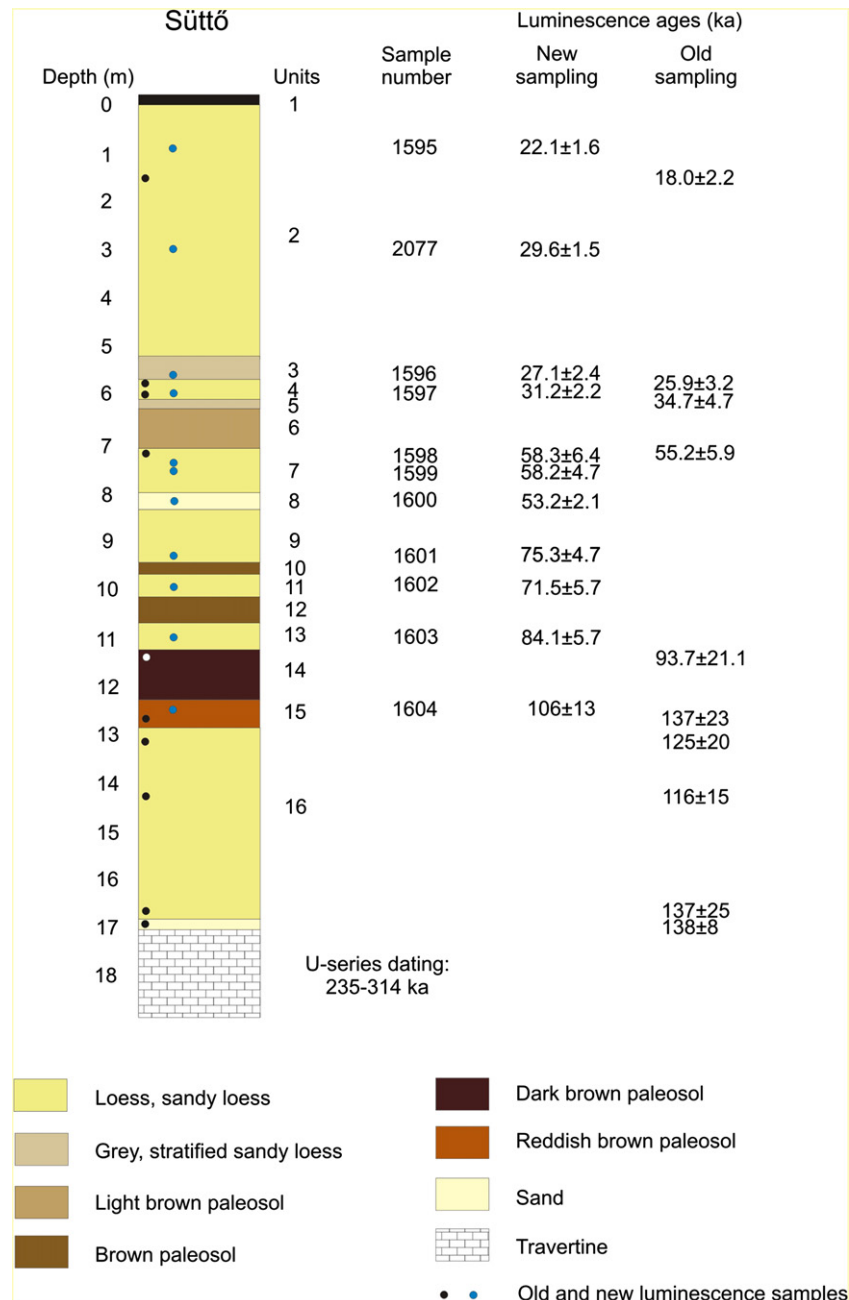


Fig. 2. Stratigraphic logs of the loess profile at Süttő, showing major sedimentological units, luminescence sample positions and (old and new) luminescence ages.

dispersion in 10 ml 1% ammonium hydroxide for at least 12 h on a rotator. Forty different samples throughout the profile, representing the three different sediment types (loess, sand and paleosol) were selected for test measurements to optimize the standard operating method. Around 250 test measurements were carried out to adjust the optimal weight of the samples and the optimal settings for the different types of sediments. Reproducibility tests were also conducted on the test samples to provide quality assurance for the measurements. During these tests, three subsamples of each sample were measured and compared. The measurement protocol was accepted if their coefficient of variation was smaller than 5%. Each measurement was repeated three to six times in order to check the reliability of the measured values. The average of the measurements was used for the calculation of the parameters of interest.

2.3. Magnetic susceptibility

Sampling for the magnetic investigations was carried out using a more sophisticated technique which yields also well oriented samples suitable not only for alternating field but also for thermal demagnetization (Schnepp et al., 2008). Low field susceptibility measurements have been done using AGICO “KLF3 Minikappa”. KLF3 is designed for laboratory measurements of magnetic susceptibility of cubic, cylindrical and fragment rock specimens. The nominal volume of the KLF3 is 10 cm³; measurements have to be recalculated to this volume. Alternatively for fragment rocks values can be normalized to bulk. Our measurements have been recalculated to 10 cm³ and normalized to mean. KLF3 works at 50 A/m magnetic field intensity and 2 kHz frequency.

The sediment pattern (soil and loess) is traced by our magnetic susceptibility record, which is sensitive to the concentration and strength of magnetic minerals. Susceptibility was used as normalizing factor for the calculation of relative paleointensity (RPI).

2.4. Luminescence dating

Luminescence is a trapped charge dating method, based on the time-dependent accumulation of trapped charges within crystal lattice defects due to natural ionising radiation. The time elapsed since the last exposure to sunlight, hence the deposition of the sediment, can be dated with luminescence. The actual intensity of the luminescence signal of the natural sample can be measured in the laboratory and compared to the artificially induced luminescence signals to estimate the natural absorbed radiation dose of the sample, termed equivalent dose (D_e). The luminescence age of the sample is calculated from the equivalent dose divided by the annual natural ionising radiation dose (dose rate) of the sample. Optically stimulated luminescence (OSL) is suitable for dating sediments, because the OSL signal of quartz and feldspar grains is very likely completely and rapidly reset to zero. Quartz is usually stimulated by blue light (OSL) and feldspars are stimulated by infrared radiation (IRSL) (Wintle, 2008). Different protocols can be applied to determine the D_e -s of the samples (Lian and Roberts, 2006; Wintle, 2008).

Samples from the N–NW wall of the Süttő section have previously been dated using IRSL multiple aliquot additive dose (MAAD) method (Novothny et al., 2009). However, the estimated IRSL MAAD ages are probably underestimated due to anomalous fading. The IRSL signal of feldspars suffers from anomalous fading, an unwanted, athermal loss of signal, resulting in age underestimation (Wintle, 1973; Spooner, 1994). Therefore, a more sophisticated IRSL single aliquot regeneration (SAR) protocol was applied for part of the samples to assess and correct anomalous fading using the correction method of Huntley and Lamothe (2001). Results are published in Novothny et al. (2010).

Ten new luminescence samples were taken during the field campaign in 2008, partly from the freshly opened SW wall and partly from further stratigraphically interesting positions of the N–NW wall. Ten samples were collected from loess or paleosol layers (LUM 1595–1599, LUM 1601–1604 and 2077) and one sample from an intercalated sand horizon (LUM 1600).

2.4.1. Equivalent dose determination – IRSL

The polymineral fine-grain fraction (4–11 μm) was prepared for loess samples. All samples were prepared using 0.1 N hydrochloric acid, 0.01 N sodium-oxalate and 30% hydrogen peroxide to remove carbonate, clay coating and organic matter from the samples, respectively. All preparations were conducted under subdued red light.

The same IRSL SAR protocol, fading tests and correction were performed for the new polymineral fine-grain samples as for the old samples to determine equivalent dose (D_e) values, as presented in Novothny et al. (2010). All luminescence measurements of the polymineral fine-grain samples were carried out at the Leibniz Institute for Applied Geophysics, in Hannover, Germany, using an automated Risø TL/OSL-DA-15 reader. IRSL stimulation was performed for 300 s at 50 °C. D_e values of the samples were obtained by integrating the 2.5–10 s regions of the IRSL decay curves and subtracting the background, the final 10 s of the stimulations. Preheat of 220 °C or 240 °C for 10 s was used before IR stimulations. The reliability of the applied protocol was checked using dose recovery test. The dose recovery ratios range from 0.96 to 1.1, which is acceptable.

2.4.2. Equivalent dose determination – OSL

The coarse grain quartz fraction (150–200 μm) was separated for the sand sample. This sample was also prepared using 0.1 N hydrochloric acid, 0.01 N sodium-oxalate and 30% hydrogen peroxide to remove carbonate, clay coating and organic matter from the samples, respectively. Quartz grains were extracted by heavy liquid separation using sodium polytungstate, followed by hydrofluoric acid (HF) etching to remove the outer part of the quartz grains and any remaining feldspar minerals from the sample.

The quartz sample was measured at the Department of Radioisotopes, Institute of Physics, Silesian University of Technology in Gliwice, Poland, using a Daybreak OSL/TL reader. Owing to the different reader, slightly different measuring conditions were adopted for the coarse-grained quartz sample. OSL stimulation was performed for 30 s at 160 °C with blue diodes. D_e values from the quartz fraction of the sample was obtained by integrating the 0–1 s region of the OSL decay curves and subtracting the background, the final 5 s of the stimulations. Preheat of 260 °C for 10 s and cutheat of 220 °C were used during measurement. Reliability of the applied protocol was checked using a dose recovery test, which yielded a recovery ratio of 0.91.

2.4.3. Dose rate determination

Dose rates were obtained from the potassium, uranium and thorium content, as measured by gamma spectrometry in the laboratory at the Leibniz Institute for Applied Geophysics, in Hannover. Dose rate conversion is based upon the factors of Adamiec and Aitken (1998). Details concerning the applied protocol, measurement properties and dose rate determination are explained in Novothny et al. (2010).

3. Results

3.1. Litho and pedostratigraphy

Seven different lithological units could be distinguished at the SW wall. The thickness of the sediment succession is 7.60 m, starting with the modern soil at the top and finishes with the loess layer beneath the uppermost brown paleosol. A part of the loess sequence (Units 3–7) is investigated at both walls of the quarry. The sediment succession of the N–NW wall has a thickness of 10.65 m. The following profile is exposed (Fig. 2) (description from the top to the bottom):

Units:

- 0.00–0.65 m: A, B and C horizons of the modern chernozem soil, intermingled by many crotonas in the upper part. The chernozem has a dark brown colour, which is gradually getting lighter to the bottom, as it has a gradual transition towards the underlying loess.
- 0.66–5.29 m: Pale yellow to light brown calcareous sandy loess. Carbonate accumulations are characteristic around root channels. Fe- and Mn nodules appear between 1.40 m below surface and the bottom of the unit, increasing in number and getting stronger towards the lower border of this horizon. Some molluscs are found in the middle of the unit.
- 5.30–5.65 m: Greyish, pale yellow, laminated sandy loess. Fe- and Mn concretions are common in former root channels in the lower part of this unit.
- 5.66–5.96 m: Pale yellow, greyish yellow sandy loess. Fe- and Mn accumulations, 1–3 mm in diameter are common along former root channels. Fe concretions are abundant in the middle of the layer, but they are absent in the lower part.
- 5.97–6.26 m: Greyish, pale yellow, laminated sandy loess, thickening in the direction of the N–NW wall. Fe- and Mn dots

occur sparsely, mostly along former root channels. It has a sharp boundary to the soil below.

6. 6.27–7.14 m: Truncated brown (reddish brown to dark brown) paleosol with strong granular structure. It has a gradual transition towards the underlying loess, but the boundary is strongly bioturbated. Its thickness varies horizontally, the thickest part is in the corner of the two walls and it is thinning out laterally in both directions. Mn nodules seldom occur in the paleosol. Grey, gleyed features 1–3 cm in diameter are characteristic in the uppermost part of the layer. Small (1–10 mm in diameter) charcoal pieces are abundant in the middle part of the soil and molluscs are common in the upper part of the unit.
7. 7.15–7.84 m: Light brown to yellow calcareous sandy loess. This layer is abundant in secondary carbonates – carbonate coatings – infilling in root channels. Only small concentrations of Mn are present along the remains of former root channels.
8. 7.85–8.10 m: Yellowish grey sand.
9. 8.11–9.28 m: Greyish, pale yellow to brownish yellow homogeneous loess, which gradually becomes lighter to the middle part of the layer and then subsequently darker, brownish to the bottom. The boundary towards the underlying soil is very sharp, a discontinuity is most likely. Remnants of former root channels are filled with carbonate.
10. 9.29–9.74 m: Truncated blackish, greyish brown paleosol with granular structure. In contrast to the abrupt transition between the overlying loess and the soil, the transition towards the underlying loess layer is gradual. Small pieces of charcoal 1–3 mm in diameter are characteristic in the entire paleosol.
11. 9.75–9.99 m: Brownish yellow (yellowish light brown) loess, which very likely can be considered as a transitional layer between the over and underlying paleosols. Its colour is closer to the colour of the underlying soil, indicating that the soil formation had not stopped abruptly and the biological activity was quite strong.
12. 10.00–10.64 m: Bioturbated yellowish brown paleosol with granular structure. Carbonate concretions 1–3 mm in diameter are common.
13. 10.65–11.09 m: Yellowish brown transitional loess layer between the overlying paleosol and the underlying pedocomplex.
14. 11.10–12.33 m: Well-developed, dark (blackish) brown, clayey chernozem-like paleosol with a strong prismatic structure. This is the upper constituent of the pedocomplex. Carbonate coatings along biogalleries, cracks and Mn nodules are common. Small crotonas 1–3 cm in diameter, filled with reddish-brown material are characteristic. The colour of the soil gradually becomes more reddish downwards, as it has a gradual transition towards the underlying loess.
15. 12.34–12.73 m: Well-developed, truncated, reddish brown, clayey paleosol, mostly preserved only in a paleochannel. The thickness of this paleosol varies horizontally and even completely vanishes in some parts of the outcrop. Carbonate coatings and Mn concretions occur along former root channels. Carbonate concretions 1–3 cm in diameter are common. It has a sharp boundary towards the underlying loess.
16. 12.74–15.00 m: Reddish yellow to pale yellow sandy loess. The top of the layer is bioturbated. Mn nodules are absent in the uppermost part of the layer, but appear downwards. Smaller carbonate concretions and loess “dolls” 2–5 cm in diameter are characteristic.

3.2. Grain size

The shape of the differential grain-size distribution curves of the Süttő profile is typical for loess–paleosol sediments. The curves represent polymodal distributions with a pronounced

shifting towards coarser grain sizes (Fig. 3). Coarse silts and fine sands dominate the grain-size spectra. The sand content ($>63 \mu\text{m}$) ranges from about 10% to about 40%. The sand content of the loess and paleosol layers slightly increase with depth. A high sand content (40%) is characteristic for the lowermost part of the profile (Unit 16) and directly beneath the sand horizon (Unit 9) and for the grey, laminated horizons (Units 3 and 5) (Fig. 4). The intercalated sand horizon (Unit 8) has almost 80% sand content. Clay content ($<4.6 \mu\text{m}$) of the loess layers is between 7 and 15%. About 15–20% clay contents are determined for the paleosols. The highest clay content was measured for the modern soil. The sand horizon in the middle of the profile has the least clay content of 4–6%.

The grain-size distribution curves of loess sediments reflect polymodal distributions, possibly constituted from three different components (Fig. 3). Beside the general bimodal distribution of loess sediments (a pronounced peak in the coarse silt fraction and a shoulder in the clay-, fine silt fraction) a third component appeared in the curves as a clear shoulder in the sand fraction. Strong bimodal distributions (twin peaks) within the silt fractions (Machalett et al., 2008) were not detected for the samples from Süttő. In some cases extra peaks occurred in the coarse sand fraction, centred at around 400 and 900 μm . This feature is common for the loess at the bottom of the profile (Unit 16) and directly beneath the sand horizon (Unit 9) and in the upper grey, laminated horizon (Unit 3), which are characterised by aforementioned higher sand content ($>63 \mu\text{m}$) and higher coarse sand content ($>160 \mu\text{m}$) (Fig. 4). The grain-size distribution curves of the paleosols are similar to those of the loess. However, the paleosols have a more pronounced clay- and fine silt component (Fig. 3). Well-sorted sand samples are illustrated by polymodal distributions with an irrelevant clay-fine silt component, a narrow, pronounced coarse silt-sand peak centred at around 100–150 μm and an additional coarse sand component between 400 and 600 μm .

A continuous, high-resolution grain-size record was obtained for the almost 16 m thick profile. Sand content ($>63 \mu\text{m}$), coarse sand content ($>160 \mu\text{m}$), clay content ($<4.6 \mu\text{m}$) and U ratio ($16\text{--}44 \mu\text{m}/5.5\text{--}16 \mu\text{m}$) (Vandenberghe et al., 1993) (Fig. 4) were calculated for interpreting the results and making comparison with other studies regarding grain-size analysis to get information concerning regional differences and similarities within and the Carpathian Basin and elsewhere.

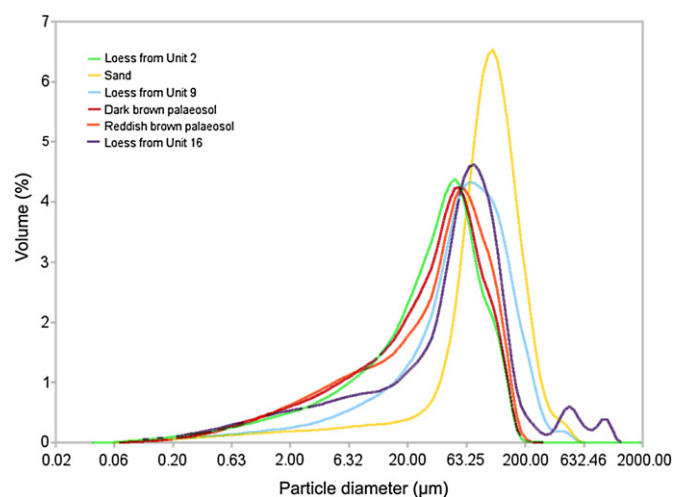


Fig. 3. Differential grain-size distribution curves from the main sedimentological units of the Süttő profile: loess (from unit 2, 9 and 16), black (chernozem-like) soil, reddish soil and sand.

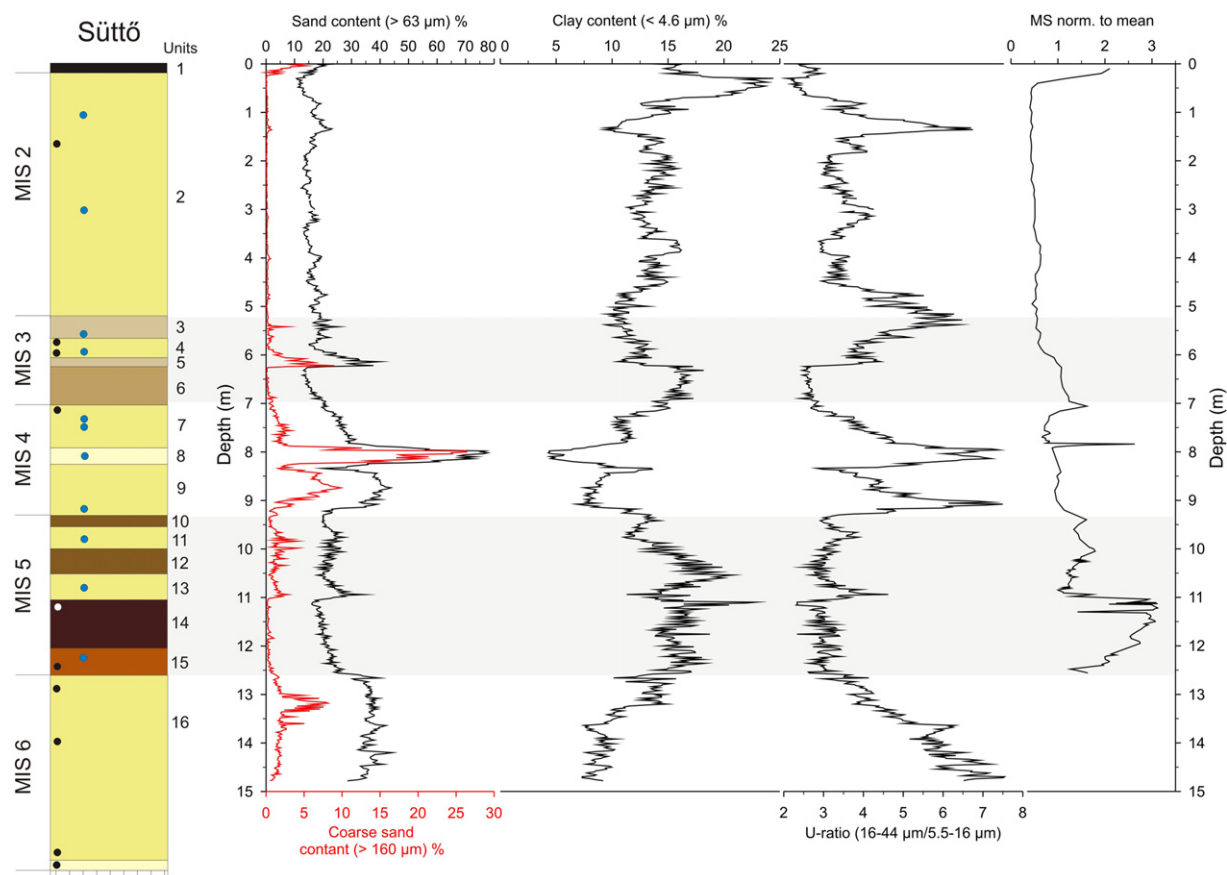


Fig. 4. Stratigraphic logs of the loess profile at Sütő, showing major sedimentological units and luminescence sample positions. Variation of grain-size parameters clay-, total sand and coarse sand contents, U ratio and magnetic susceptibility curves are shown along the log of the profile.

3.3. Magnetic susceptibility

Magnetic susceptibility values are between 186×10^{-6} [SI] and 1350×10^{-6} [SI] along the profile (Fig. 4; values normalized to mean). Lowest values are typical for the loess layers at the uppermost part of the profile (Unit 2). Slightly higher MS values are characteristic for the loess in the middle part of the composite profile (Units 7, 9). The sand horizon (Unit 8) could not be entirely sampled for magnetic susceptibility measurements, due to its loose sandy texture. Buried soil layers often show higher MS values, which is interpreted as a consequence of higher precipitation, due to in-situ formation of magnetizable minerals as a function of precipitation. Highest values are characteristic for the modern soil on the top of the profile (Unit 1) and for the pedocomplex (Units 14, 15). There is a slight difference between the upper and lower part of the pedocomplex. Lower MS values are typical for the brownish buried soils in the middle of the profile (Units 6, 10, 12). The MS values of the grey laminated horizons (Units 3, 5) are identical to the bracketing loess or paleosol layers.

3.4. Luminescence dating

The measured concentrations of uranium, thorium and potassium content in the samples along with cosmic dose rates and total dose rates are summarized in Table 1. Total dose rate values range from 2.71 ± 0.15 to 3.57 ± 0.19 Gy/ka for polymineral fine-grains and is 1.94 ± 0.07 Gy/ka for the coarse grain quartz. D_e values range from 48.8 ± 0.3 to 325.5 ± 2.6 Gy (Table 2). Fading rates (g_{2days} -value), calculated from the initial part (0–1 s) of the IRSL decay

curves are between 2.4 and 4.0%/decade. These values are similar to the reported fading rates from other loess records (Buylaert et al., 2007, 2008; Thomsen et al., 2008), whereas fading rates calculated from the middle part (2.5–10 s) of the IRSL decay curves range from 1.4 to 2.5%/decade. Therefore, D_e determination, fading rate and age calculations were based on calculations from the middle part of the IRSL decay curve (see Novothny et al., 2010).

Previously published IRSL MAAD ages of the Sütő profile (Novothny et al., 2009) had been calculated without fading correction. They are very likely underestimated due to anomalous fading. The new, fading corrected IRSL SAR ages (Table 2) range from 22.1 ± 1.6 (LUM 1595) to 105 ± 13 ka (LUM 1604). They are 14–25% higher than the corresponding uncorrected IRSL SAR ages.

Table 1

A summary of uranium, thorium and potassium content, as determined by gamma spectrometry in the laboratory and the cosmic dose rate.

Sample number	Uranium [ppm]	Thorium [ppm]	Potassium [%]	Cosmic dose rate [mGy/a]
1595	2.55 ± 0.02	8.40 ± 0.05	1.09 ± 0.01	0.193 ± 0.019
2077	3.23 ± 0.03	10.92 ± 0.06	1.44 ± 0.01	0.148 ± 0.015
1596	3.02 ± 0.02	10.36 ± 0.06	1.32 ± 0.01	0.109 ± 0.011
1597	3.04 ± 0.02	10.13 ± 0.06	1.28 ± 0.01	0.104 ± 0.010
1598	3.08 ± 0.02	10.73 ± 0.06	1.40 ± 0.01	0.088 ± 0.009
1599	3.08 ± 0.01	10.20 ± 0.04	1.26 ± 0.01	0.079 ± 0.008
1600	2.89 ± 0.02	10.57 ± 0.06	1.00 ± 0.01	0.074 ± 0.007
1601	3.19 ± 0.02	10.27 ± 0.06	1.25 ± 0.01	0.066 ± 0.007
1602	2.94 ± 0.02	9.93 ± 0.05	1.25 ± 0.01	0.062 ± 0.006
1603	2.82 ± 0.02	9.72 ± 0.06	1.09 ± 0.01	0.056 ± 0.006
1604	3.24 ± 0.02	13.55 ± 0.06	1.45 ± 0.01	0.050 ± 0.005

Table 2

Dose rates, equivalent doses, fading rates, uncorrected and fading corrected IRSL ages and OSL ages of the samples.

Sample number	Dose rate [Gy/ka]	Equivalent doses [Gy]	Uncorrected IRSL age [ka] 2.5–10 s	g_2 days-value [%/decade]	Corrected IRSL age [ka] 2.5–10 s	Quartz OSL age [ka]
1595	2.71 ± 0.15	48.77 ± 0.29	18.0 ± 1.0	2.3 ± 0.1	22.1 ± 1.6	
2077	3.38 ± 0.13	76.06 ± 0.57	22.5 ± 0.9	2.5 ± 0.1	29.6 ± 1.5	
1596	3.16 ± 0.17	71.91 ± 0.60	22.8 ± 1.3	1.9 ± 0.1	27.1 ± 2.3	
1597	3.10 ± 0.17	78.20 ± 0.64	25.2 ± 1.4	2.3 ± 0.1	31.2 ± 2.2	
1598	3.26 ± 0.18	162.50 ± 2.06	49.9 ± 2.8	1.7 ± 0.2	58.3 ± 6.4	
1599	3.08 ± 0.17	154.64 ± 1.75	50.2 ± 2.9	1.6 ± 0.1	58.2 ± 4.7	
1600	1.94 ± 0.07	103.28 ± 0.86				53.2 ± 2.1
1601	3.10 ± 0.18	205.19 ± 4.15	66.2 ± 4.0	1.4 ± 0.0	75.3 ± 4.7	
1602	2.99 ± 0.17	184.69 ± 2.22	61.9 ± 3.6	1.7 ± 0.1	71.4 ± 5.7	
1603	2.78 ± 0.16	198.67 ± 2.44	71.4 ± 4.2	1.7 ± 0.1	84.1 ± 5.7	
1604	3.57 ± 0.19	325.53 ± 2.61	91.1 ± 4.9	1.6 ± 0.2	105.5 ± 13.1	

The uppermost part of the composite profile consists of loess layers. The upper grey, laminated horizon deposited between 22.1 ± 1.6 ka (LUM 1595) and 27.1 ± 2.4 ka (LUM 1596) or 29.6 ± 1.5 ka (LUM 2077). The uppermost brownish paleosol and the covering lower, grey laminated horizon developed between 31.2 ± 2.2 ka (LUM 1597) and 58.3 ± 6.4 ka (LUM 1598), measured from loess, bracketing these layers. Samples LUM 1598 and LUM 1599 were collected from the same loess layer underlying the uppermost paleosol (Unit 7), but from two different walls of the quarry, to check the reliability of the presumed linkage of the whole, composite profile. Fading corrected IRSL SAR ages of these samples (58.3 ± 6.4 ka and 58.2 ± 4.6 ka) show excellent agreement, verifying the presumed linkage between the two profiles. Fading corrected IRSL ages are supported by the OSL SAR age (53.2 ± 2.1 ka) of the coarse-grained quartz sample from the sand layer (LUM 1600), which is slightly younger compared to the fading corrected IRSL SAR ages from the overlying loess layers (58.2 ± 4.6 ka), however, they are in agreement within the error margins. Two thinner brownish paleosols in the middle of the composite profile formed between 75.3 ± 4.7 ka (LUM 1601) and 84.1 ± 5.7 ka (LUM 1603), as determined from the loess layers bracketing the two paleosols. The lower part of the pedocomplex yielded IRSL SAR age of 105 ± 13 ka (LUM 1604).

The underlying loess layers were not sampled for this study, as they had already been investigated by Novothny et al. (2010). The latter study concludes, that though the fading corrected IRSL SAR ages and radiocarbon ages from the upper part of the profile are in good agreement, fading corrected IRSL SAR ages obtained from the lowermost part of the profile still seem to be underestimated, compared to the Infrared radiofluorescence (IRRF) ages. Alternatively the IR-RF age overestimates the true deposition age owing to insufficient bleaching. Therefore, fading corrected IRSL SAR ages from the penultimate glacial loess can be considered as minimum ages (Novothny et al., 2010).

4. Discussion

Detailed investigation of grain-size distributions and magnetic susceptibility of windblown sediments is a useful approach to assess the triggering, prevailing climatic factors for the time of deposition. Grain-size distribution of loess is mainly a function of wind intensity (speed, strength) and distance from the main sediment source. In general, coarser grains accumulate during colder periods, when wind intensity was stronger, and finer grains deposited in warmer periods, when climatic conditions were not in favour for stronger winds. Dominance of finer aeolian dust deposition during warmer periods and coarser dust in cold periods is confirmed by the determination of the U ratio ($16\text{--}44 \mu\text{m}/5.5\text{--}16 \mu\text{m}$) (Vandenberghe et al., 1985). Higher U ratios imply

colder periods and lower U ratios indicate warmer climate. The advantage of the U ratio is that it ignores the secondary formed clay minerals and sand size grains, which deposited by saltation (Vandenberghe et al., 1997). However, the sand content also provides important information about predominating wind regime and short events triggered by abrupt increases of aeolian dynamics. Clay content of the samples is a result of pedogenetical processes, therefore it is considered as a good marker for warmer and more humid periods with soil formation.

The chronostratigraphical interpretation is discussed referring to the marine isotope stages from MIS 6 to MIS 2 (Martinson et al., 1987) to make the correlation easier. The formation of the travertine correlates to MIS 7–9, as determined by uranium-series dating yielding $^{230}\text{Th}/\text{U}$ ages ranging from 240 to 310 ka (Sierralta et al., 2010).

4.1. MIS 6

Penultimate glacial loess (Unit 16) between 15 and 12.74 m total depth (Figs. 2 and 4) accumulated on the sand-gravel bed, deposited probably by the small Bikol Creek west of the quarry, overlying the travertine. Age determination of the loess layer was obtained by luminescence dating yielding minimum ages (Novothny et al., 2010). AAR measurements support the formation of this loess layer during MIS 6 (Novothny et al., 2009). Unfortunately, it is not clear based only on the available chronological dataset whether the loess sequence represents the whole isotope stage, or (very likely) only a part of it owing to remarkable erosion.

MS dataset is not available from this part of the profile, though previous MS measurements, carried out using a portable Kappa-meter KT-5, reported the MS values to be low as in the uppermost part of the profile (Novothny et al., 2009). Low MS values are linked to drier climate and indicate the absence of any soil forming process. This is supported by the grain-size data, as the lowest clay content ($<4.6 \mu\text{m}$) of the whole profile was determined for the penultimate glacial loess. Two distinct horizons are detected within this loess unit based on the clay content. The average clay content of the upper horizon ranges from 12.5 to 15% and those of the lower horizon are between 7.5 and 10% (Fig. 4). The higher and increasing clay content of the upper horizon is very likely related to the overlying pedocomplex, since the transition of this layer towards the buried soils is strongly bioturbated. In contrast to the clay content, the sand content ($>63 \mu\text{m}$) of this loess unit is very high, ranging from 32 to 42%. These values are higher than those determined for the upper part of the loess sequence. Similarly to the total sand content, the coarse sand content ($>160 \mu\text{m}$) of this unit is also very high, ranging from 1 to 2%, but increasing upwards through the profile. There is a local peak around 13.2 m in total depth, with a coarse sand content of 7.5%. The grain-size distribution of loess

deposits is a good indicator for changes in wind regime (Vandenberghe et al., 1985, 1997, 1998; Vandenberghe and Nugteren, 2001; Sun et al., 2002). Therefore, this high content of sand, especially coarse sand might imply different paleo-wind conditions during the penultimate glacial cycle compared to the Late Pleniglacial at Süttő. More intensive sedimentation of coarser grains can be attributed to changes in wind regime or changes in available source material from the fluvial system of the River Danube (Marković et al., 2008). Presumably the source material from the Danube floodplain has not been changed significantly, neither due to the variation in the fluvial regime of the River Danube nor due to the altering wind direction. However, the source material is partly originating from the small Bikol Creek west of the quarry and significant long term changes of its material can be assumed due to landscape alterations of the nearby Gerecse Hills. Therefore, the most likely explanation for the changes can be imputed to the altering source material available from Bikol Creek, besides altering wind intensity. Increased wind intensity is assumed during the penultimate glacial cycle with cold, but gradually warming climate towards the approaching interglacial. The latter assumption is confirmed by the U ratio of this loess unit, showing a gradual decrease towards the overlying pedocomplex.

4.2. MIS 5

Three warm (5a, c, e) and two colder (5b, d) periods are distinguished during isotope stage 5. Unfortunately, sediments from MIS 5 cannot be considered complete from most of the important loess sections in Hungary (Frechen et al., 1997; Novothny et al., 2002). However, at Süttő, a former paleo-channel, incised into the penultimate glacial loess and preserved the sediments from these periods. A very detailed MIS 5 paleosol-loess sequence can be examined in the nearby area of the paleo-channel. Over 10 m in both directions of the paleo-channel, the paleosols thin out and merge, and the older soils are superimposed by younger ones. Sediments were examined a few metres away from the paleo-channel, where five different units were distinguished between 12.73 and 9.29 m total depth. The MIS 5 sequence starts with a well-developed, reddish-brown paleosol (Unit 15) covered by a well-developed dark brown chernozem-like paleosol (Unit 14). A similar sequence for paleosols correlating with MIS 5 is reported from the Irig and Ruma sections in Serbia by Marković et al. (2004, 2007) and from Ukraine by Łanczont and Boguczkyj (2007). At the Süttő section the pedocomplex is covered by a transitional loess layer (Unit 13) and two brownish paleosols (Units 12 and 10) intercalated by transitional loess layer (Unit 11). Samples for luminescence dating were taken from both units of the pedocomplex (LUM 550, LUM 1604, LUM 549) and from both transitional loess layers (LUM 1602, LUM 1603) to provide a chronological framework for the sediments from MIS 5. Luminescence ages from the reddish-brown paleosol (105 ± 13 ka for sample LUM 1604 and 137 ± 23 ka for sample 549; see the latter in: Novothny et al., 2010) suggest that this soil probably formed during the last interglacial period (MIS 5e). The overlying chernozem-like paleosol has an IRSL age of 93.7 ± 21.1 ka (LUM 550). Considering the big error ranges, the time of the soil development cannot be deduced properly based only on the IRSL age of this sample. However, based on the bracketing IRSL ages it is likely that the chernozem-like soil is the result of the warm and more humid period of MIS 5c. The transitional loess layer (Unit 13) overlying the pedocomplex yielded IRSL age of 84.1 ± 5.7 ka (LUM 1603), showing a likely correlation with MI substage 5b. Two thinner brownish steppe-like soils and an intercalated transitional loess layer indicate a shorter and/or less warm and humid interstadial period interrupted by a colder period resulted in thin loess

sedimentation during MI substage 5a. This theory is supported by IRSL ages determined from the bracketing loess layers (LUM 1603–1601), which yielded ages of 84.1 ± 5.7 ka, 71.5 ± 5.7 ka and 75.3 ± 4.7 ka, respectively (Fig. 2).

The MS curve of these units confirms the aforementioned stratigraphy. Highest MS values are recorded within the pedocomplex, showing strong soil formation (Fig. 4). Strong correlation between MS values and paleo-precipitation was suggested by Heller and Evans (1995), Maher et al. (2002, 2003) and Nugteren et al. (2004), although it can not be clearly detected within the pedocomplex at Süttő. Based on the preliminary results of soil micromorphology, the lower, reddish-brown soil is supposed to have developed under wetter climate, while the upper chernozem-like soil is a good marker of a warm, but drier climate with steppe-like vegetation. The pattern of the MS curve does not reflect this hypothesis, because the highest values are detected in the upper part of the chernozem-like soil and a gradual decrease is shown in the rest of the pedocomplex towards the underlying loess layer. An abrupt decrease in MS values is observed in the loess (Unit 13) overlying the pedocomplex. However, the MS signal is not as low as in a typical loess layer, thus giving evidence for the transitional character of this unit. Higher MS values were obtained from the brown paleosols most likely formed during MIS 5a, though these signals are only the half of the MS signals detected in the pedocomplex.

The clay content of sediments deposited during MIS 5 differ depending on the type of sediments (Fig. 4). Buried soils are enriched in clay due to soil forming processes. Similar high clay content is shown from the pedocomplex (Units 14 and 15) and the lower brown paleosol (Unit 12), ranging between 13 and 23%. Two peaks in the clay content are observed within the pedocomplex, showing evidence for two periods of soil formation. The upper brown steppe-like paleosol (Unit 10) has a lower clay content ranging from 11 to 14%, indicating a strong, but rapid soil forming period.

The sand content is lower in the buried soils ranging from 15 to 20% and higher in the transitional loess layers with values around 25%. The total sand content is higher in the lower part of the pedocomplex (30–40%), after it a gradually decreasing trend is observed during MIS 5e–5c, followed by a slight increase during MIS 5a–b (Fig. 4). This unusually high contribution of sand-sized grains is very likely owing to the vicinity of the source material in the Danube floodplain. Pattern of coarse sand content follows those of total sand content, although the contribution of coarse sands was higher during MIS 5b–a.

Calculated U ratios are between 2.5 and 3 in the buried soil layers and between 3 and 4 in the transitional loess units. However a strong decrease in the U ratio is observed during MIS 5. This is not the only characteristic decrease of the U ratio within the entire profile at Süttő, in contrast with the general trends in loess–paleosol sequences, for example at Surduk in Serbia (Antoine et al., 2009), at Remisowka in Kazakhstan (Machalett et al., 2008), or in the Chinese loess (Vandenberghe et al., 1997; Nugteren et al., 2004). The reason could be the relative abundance of coarser material in this part of the profile, which results in much higher U ratios (Nugteren et al., 2004).

4.3. MIS 4

Lower Pleniglacial loess deposited on the Early Glacial paleosols between 9.28 and 7.15 m depth, which is confirmed by IRSL ages of sample LUM 1598–1601 (58.3 ± 6.4 ka 58.2 ± 4.6 ka 75.3 ± 4.7 ka, respectively) obtained from the loess layer of Units 8 and 10 and an OSL age of sample LUM 1600 (53.2 ± 2.1 ka) determined from the sand horizon (Unit 9) (Fig. 2).

The MS curve of this part of the profile is showing lower values than the sediments from MIS 5, which is characteristic for loessic sediments. Directly above, the intercalated sand horizon at 7.8 m depth yielded very high MS values (Fig. 4), which requires further magnetic investigations.

The sediment sequence deposited during MIS 4 can be divided into three, distinct parts by means of sedimentology and magnetic susceptibility. These three units are equivalent with the previously described sedimentological units (Units 10, 9 and 8). After the mild and more humid climate of the Early Glacial, the climate of the Lower Pleniglacial probably became colder, drier with increased wind intensity. This stronger wind intensity resulted in deposition of coarser grains in higher proportions, as seen for Units 10 and 9. High total sand content (40%), very high coarse sand content (7–10%) and low clay content (7–9%) of loess from Unit 10 indicate some kind of similarity to those of penultimate glacial loess. Therefore, similar climate conditions are presumed during the beginning of MIS 4 and MIS 6. Very likely a storm event resulted in the deposition of a sand horizon in the middle of the profile (Unit 9) with 80% sand content, 25% coarse sand content and very low clay content (5%). This was probably followed by an also cold, but less windy period at the end of MIS 4 based on the results of the grain-size measurements of the covering loess unit. Total sand and coarse sand content of the loess Unit 8 are still high (30% and 2–3%, respectively), but not as much as in the previously accumulated sediments. The first two different periods and/or events are clearly observed as two distinct peaks in the curve of the U ratio between 9.3 and 7.8 m depth (Fig. 4). Similar sedimentation and cold climate with periodically occurred strong winds was reported by Antoine et al. (2009), Bokhorst et al. (2006, 2009, in press) and Marković et al. (2008) for loess from Surduk and Titel in the southern Carpathian Basin of Serbia.

4.4. MIS 3

Based on the available chronological dataset, complex sedimentation was observed during the Middle Pleniglacial at Süttő. The sediment sequence between 7.14 and 5.3 m total depth, which deposited during MIS 3 consists of four parts: a brownish paleosol (Unit 6), two greyish laminated horizons (Units 5 and 3) and an intercalated loess layer (Unit 4). The brown paleosol and the overlying laminated horizon developed between 58 and 32 ka based on the IRSL ages (58.3 ± 6.4 ka and 31.2 ± 2.2 ka) of the bracketing loess layers (LUM 1598–1597). IRSL ages determined from the overlying loess unit and from the upper laminated horizon (LUM 1597–1596) yielded 31.2 ± 2.2 ka and 27.1 ± 2.4 ka, respectively (Fig. 2). Sediments from MIS 3 were studied from both walls of the quarry. The thickness of the brown paleosol varies horizontally, its thickest part is situated in the corner of the two walls and it is laterally thinning out in both directions, which very likely reflects the paleo-relief.

A flat MS peak correlates to the brown paleosol, implying a slightly wetter and warmer climate during a warm period of MIS 3. The calculated U ratio of the paleosol (~ 2.5) yielded almost the lowest value of the entire profile with a dominance of the fine silt fraction in the grain-size fraction, suggesting a warm period. The soil is characterised by higher clay content ($\sim 15\%$) while its sand content is low and gradually decreases upwards from 25 to 12–13%. These grain-size data indicate lower wind intensity and/or limited sediment supply from the source area most likely owing to vegetated surfaces. The latter assumption is supported by the mollusc fauna of the soil (E. Krolopp, unpublished data). The mollusc fauna indicates that climate was almost as warm as today (mean July temperature: 20.9°C) with lesser precipitation during the interstadial period, which favoured steppe-forest steppe vegetation

(E. Krolopp unpublished data). Paleotemperature reconstructions showed also similar to present mean July temperatures (21.5°C) at the Irig section, in Serbia (Marković et al., 2007).

The brown paleosol is followed by two greyish, pale yellow, laminated sandy loess horizons separated by a sandy loess layer. The clay content about 13% of this loess sequence (Units 5–3) is lower than in the brown soil indicating moderate soil formation synchronous with continuous sedimentation or bioturbation. The total sand content and the coarse sand content of the sandy loess (Unit 4) are low ranging from 11 to 16% and from 0.3 to 0.5%, respectively. Two higher peaks appear on the curves of the total sand (30–40%) and coarse sand content (5–12%), indicating increased wind intensity during the deposition of the laminated horizons (Fig. 4). Similar features to these grey, stratified horizons have not yet been described from the loess sequences in the southern Carpathian Basin. However, similarities are likely between the finely laminated calcareous loess and/or tundra gley horizons from the loess sequences in Western Europe (Antoine et al., 2001, 2009), which are often developed on interstadial soils, indicating climatic deterioration prior to pure loess deposition (Antoine et al., 2001).

The MS values gradually decrease upwards from the characteristic MS value of the brown soil to those of the sandy loess units. Based on the investigation of the mollusc fauna of Units 5 and 4, generally mild climate was presumed for the rest of MIS 3 (E. Krolopp, unpublished data).

4.5. MIS 2

More homogeneous sandy loess with a thickness of 5.3 m (Unit 2) accumulated during MIS 2 at Süttő. The IRSL ages (22.1 ± 1.6 ka and 29.6 ± 1.5 ka) of samples LUM 1595 and 2077, obtained from beneath the modern soil at around 1 and 3 m depth, respectively, confirmed, that loess sedimentation at Süttő probably ceased around 18–20 ka (Novothny et al., 2009). Very high sedimentation rates (0.61 mm/y) and accumulation rates (1009 g/m²/y) can be assumed at the Süttő section for the time period between 18 and 27 ka (Fig. 2). Considering the given time range, the accumulation rate is even higher than is presented by Újvári et al. (in press), nevertheless it was among the higher ones calculated for the whole area within and around the Carpathian Basin (Újvári et al., in press; Fedorowicz and Lanczont, 2007). However, comparing this result with the accumulation rates calculated for Europe, it is among the moderate values which were determined from Western Germany (Frechen et al., 2003).

Total sand and coarse sand content of the sediments deposited during MIS 2 reflect the more or less homogeneous appearance of this loess unit, as they are steady along this part of the sequence, ranging from 12 to 20% and from 0.2 to 0.5%, respectively (Fig. 4). This part of the profile yielded the lowest values in total sand content and coarse sand content, whereas the sandy loess sequence deposited during MIS 6 and 4 gave the highest values. This grain-size distribution might indicate lower paleo-wind intensity and/or altering local sediment availability from Bikol Creek during MIS 2 than during MIS 6 or 4. Alternating wind directions within and around the Carpathian Basin can be a possible reason for the decrease in coarser material during MIS 2, as previously reported by Bokhorst et al. (2009), suggesting N–NW winds during MIS 2 but W winds during MIS 4–3.

This (quasi) homogeneous appearance of loess Unit 2 is confuted by the clay content and U ratio. They are more fragmented and allowing a recognition of three distinct (sub)cycles. Two peaks with higher clay content (12–15%) appear on the curve of the clay content centred at around 4 and 2 m depth, beside the pronounced peak caused by the modern soil (Fig. 4). At similar positions the

curve of the U ratio shows depressions, showing slightly warmer periods during MIS 2 with dominating fine silt deposition. The older warmer period is might be observed on the MS curve as well, as it contains a small, flat “peak” in the same position, at around 4 m depth. The younger warmer period is verified by the malaco-thermometer method calculated from the mollusc fauna of the loess Unit 2 between 2.5 and 0.9 m depth. Calculations were carried out by Krolopp (unpublished data) using the malaco-thermometer method as suggested by Sümegi and Krolopp (2002). Krolopp (unpublished data) reported cold (mean July temperature of 12.9 °C), but wet climate during the younger warmer period, which had turned to a bit colder climate with mean July temperature of 11.8 °C and drier conditions for the last colder period (Krolopp, unpublished data) detected in the loess sequence at Süttő. Higher mean July paleotemperatures (17–20 °C) were determined for the uppermost loess layer (V-L1L1) of the Irig section, in Serbia (Marković et al., 2007) based on the same method, which shows higher climate gradient between the southern and northern part of the Carpathian Basin during MIS 2, compared to the lower climate gradient existed during the interstadial period of MIS 3, or at present. In contrast to the recognised three (sub)cycles during MIS 2 at Süttő, the loess sequence at Surduk, in Serbia contains 4–5 cycles during the entire MIS 2 (Antoine et al., 2009).

5. Conclusion

The loess–paleosol succession at Süttő enables a detailed reconstruction of Late Pleistocene climate and environmental changes. The multi-dating approach provided an excellent chronological framework for the sediment succession at Süttő. High-resolution granulometric and paleomagnetic investigations were carried out to improve understanding regarding paleoclimate and environmental changes during the penultimate and last glacial cycles.

Fading corrected IRSL ages are in agreement with AAR and radiocarbon dating, although fading corrected IRSL ages from the penultimate glacial loess tend to underestimate the real deposition age. Therefore, the IRSL data calculated for this period has to be carefully interpreted and considered as minimum ages. Based on the presented chronological data, the loess–paleosol succession at Süttő represents a quasi-continuous sediment record for the time period ranging from MIS 6 to MIS 2 and the most detailed loess/paleosol sequence in Hungary covering an age span of at least the past 150,000 years. Four paleosol horizons formed during MIS 5. Sandy loess and sand about 2 m thick deposited during MIS 4, which is covered by a paleosol and two laminated horizons formed during MIS 3. The sediment sequence correlating to MIS 3 is covered by loess more than 5 m thick, which accumulated during MIS 2.

High-resolution grain size and magnetic susceptibility investigations were carried out from the entire loess–paleosol sequence. The MS curve of Süttő shows a clear difference between loess and paleosol horizons, indicating the alterations of arid and more humid environments. Clay-, sand-, coarse sand content and U ratio were calculated from the results of high-resolution grain-size measurements. The clay content showed excellent agreement with the MS curve, both are providing a good proxy for the local temperature and precipitation regime. Beyond that, clay content, U ratio and the MS curve indicated a detailed resolution of the climate during MIS 2. The total sand and coarse sand content showed a similar pattern like those of the loess–paleosol sequences in Serbia (Surduk, Titel) with enhanced deposition of coarser material during MIS 6–4 compared to MIS 2 (Marković et al., 2008; Antoine et al., 2009). This similarity implies regional significance of the wind dynamics and might denote changes in general wind

circulation owing to alteration of wind direction during MIS 2 (Bokhorst et al., 2009, in press) or changes in wind intensity and/or material availability upwind, which might be more relevant at Süttő. Some differences are also shown at Süttő, compared to the aforementioned sections in Serbia:

- In contrast to other investigated loess sections in the southern Carpathian Basin, higher contribution of coarser material is detected indicating the vicinity of the source material from the floodplain of the Danube and its tributaries.
- The presence of some greyish, laminated horizons in the sequence deposited during MIS 3 provide evidence for the intermediate character of the Süttő profile (or generally the loess in the northern Carpathian Basin) between the wetter climate induced diverse Western European loess sequences and the drier loess successions in the southern Carpathian Basin with total absence of tundra gleys and laminated loess layers.

Investigation of the mollusc fauna from the sediments accumulated during MIS 3 and 2 supported and supplemented our interpretation; giving evidence for a similar warm, but less wet climate during MIS 3 and revealing a sub-sequential changing of the climate during late MIS 2.

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